

DRAVAGEN, Sweden

WMO: 023170

Lat: 62.094N

Lon: 13.609E

Elev: 1857

StdP: 13.74

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.8	-11.2	-21.7	1.8	-16.6	-15.6	2.5	-11.1	22.1	32.7	19.0	30.4	2.3	250	0.691

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	74.9	59.2	71.7	58.2	68.2	56.7	62.1	70.8	60.2	67.7	58.2	65.1	6.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.8	79.3	64.2	57.0	74.1	63.0	55.2	69.4	61.1	28.7	71.2	27.4	67.8	26.0	65.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	15.3	13.1	DB	-23.1	78.7	4.0	3.3	-26.0	81.1	-28.3	83.1	-30.5	85.0	-33.4	87.4	(4)
(5)			WB	-24.9	63.9	4.2	2.4	-27.9	65.6	-30.4	67.0	-32.7	68.4	-35.8	70.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.8	16.0	17.9	23.3	33.2	43.3	50.6	55.5	52.5	44.8	35.0	25.7	18.8
	DBStd	16.07	11.78	10.72	9.72	6.73	6.70	5.54	5.16	5.07	5.63	6.89	8.80	11.95
	HDD50	5943	1055	898	829	504	227	57	11	31	170	467	728	966
	HDD65	11016	1520	1318	1294	954	671	431	296	388	605	930	1178	1431
	CDD50	402	0	0	0	0	21	77	181	108	14	1	0	0
	CDD65	2	0	0	0	0	0	0	2	0	0	0	0	0
	CDH74	124	0	0	0	0	5	24	83	12	0	0	0	0
	CDH80	5	0	0	0	0	0	1	4	0	0	0	0	0
Wind	WSAvg	5.4	5.3	5.5	6.3	5.9	6.1	6.0	5.0	4.6	5.1	5.3	4.9	5.1
Precipitation	PrecAvg	23.8	1.2	1.0	1.0	1.3	1.9	2.9	3.5	3.3	2.4	2.1	1.7	1.4
	PrecMax	30.7	2.2	2.0	1.9	2.8	3.8	5.9	6.2	6.4	4.4	4.8	4.6	2.1
	PrecMin	18.6	0.2	0.3	0.2	0.3	0.4	0.7	1.9	1.7	1.1	0.5	0.7	0.4
	PrecStd	3.4	0.6	0.5	0.5	0.7	0.9	1.2	1.1	1.3	1.1	1.2	0.9	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.8	41.0	47.2	59.5	74.3	77.2	80.5	75.8	66.4	57.1	48.0	41.3
		MCWB	36.4	35.9	39.2	46.9	54.9	57.2	63.0	60.6	54.8	50.8	45.6	38.6
	2%	DB	37.5	38.6	43.1	54.4	68.9	72.7	76.3	71.8	62.6	52.2	43.5	39.2
		MCWB	35.0	35.1	37.4	43.5	53.7	56.3	61.5	59.4	53.8	47.5	41.6	36.7
	5%	DB	35.1	36.2	39.8	50.0	63.6	68.5	72.5	68.1	58.9	49.3	40.0	37.2
		MCWB	33.1	33.4	35.3	41.6	51.1	55.2	59.8	57.5	52.5	45.3	38.5	35.0
	10%	DB	32.3	33.2	37.3	45.6	58.0	64.3	68.8	64.2	55.7	46.5	37.3	34.7
		MCWB	30.5	31.1	33.6	38.8	48.4	53.0	58.1	56.4	50.5	44.0	35.8	32.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.9	37.4	40.2	48.6	56.8	61.2	64.8	63.4	57.5	52.1	45.7	39.2
		MCDB	39.1	39.9	45.9	56.8	68.2	69.6	74.9	70.4	62.6	55.2	47.6	40.6
	2%	WB	35.4	35.4	37.9	44.7	54.7	58.3	62.9	61.2	55.3	48.3	41.6	36.9
		MCDB	37.3	38.0	42.2	53.1	65.7	67.2	73.1	67.7	60.6	51.5	43.3	38.6
	5%	WB	33.4	33.5	35.7	41.9	52.2	56.3	61.2	59.3	53.2	46.0	38.7	35.1
		MCDB	35.2	36.1	39.5	48.6	61.3	66.1	70.1	65.4	57.8	48.2	39.8	36.9
	10%	WB	30.6	31.2	33.8	39.5	49.3	54.4	59.4	57.4	51.3	44.1	35.9	33.0
		MCDB	32.2	33.3	36.7	44.8	57.4	63.0	66.8	62.9	55.2	46.7	37.0	34.5
Mean Daily Temperature Range	5% DB	MDBR	15.9	17.3	18.8	17.9	19.2	20.1	20.0	19.2	17.2	14.1	12.0	14.9
		MCDBR	14.3	15.4	19.0	25.7	31.0	29.6	28.5	26.7	24.1	19.3	13.5	13.5
		MCWBR	13.0	12.7	14.1	15.5	16.6	15.0	14.6	15.0	15.8	14.4	12.0	12.2
	5% WB	MCDBR	13.5	13.8	17.0	23.6	27.1	24.9	25.6	21.8	21.0	16.4	12.7	13.4
		MCWBR	12.4	11.8	12.7	14.7	15.3	13.6	14.1	13.3	14.6	13.1	11.5	12.3
Clear-Sky Solar Irradiance	taub	0.204	0.257	0.285	0.318	0.328	0.324	0.339	0.329	0.299	0.284	0.219	N/A	
	taud	2.114	2.281	2.308	2.374	2.477	2.516	2.483	2.500	2.567	2.499	2.192	N/A	
	Ebn at Noon	165	231	265	278	284	286	278	272	262	223	155	N/A	
	Edh at Noon	10	19	26	30	30	30	30	27	21	16	9	N/A	
All-Sky Solar Radiation	RadAvg	79	279	760	1308	1499	1631	1488	1151	739	326	100	41	
	RadStd	9	48	69	116	127	138	185	104	86	51	21	4	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page